

# Amine Hamlouchi

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## EDUCATION

**University of Louisville, J.B. Speed School of Engineering**

*Bachelor of Arts in Computer Science; Minor in Arabic; GPA: 3.5/4.0*

Louisville, KY

*Expected May 2028*

## TECHNICAL SKILLS

**Languages:** Python, Java, C, C++, JavaScript/TypeScript, SQL, Bash, HTML/CSS

**AI/ML & Data:** RAG (retrieval-augmented generation), LLM orchestration, prompt engineering, NLP / information extraction, ETL pipelines, pandas, NumPy, scikit-learn, model evaluation

**Web & Mobile:** React, React Native, Next.js, Node.js/Express, FastAPI, REST APIs, PostgreSQL/PostGIS, Supabase

**Infrastructure:** Docker/Podman, GCP Cloud Run, Vercel, Pulumi (IaC), GitHub Actions (CI/CD), Firebase Auth, Stripe, Git, Linux

## EXPERIENCE

### Product Engineering Intern

Aug. 2026 – Present

*Kamel Ride*

*Remote*

- Ship features and fixes in a production monorepo (Node/Express with PostgreSQL + PostGIS, React/TypeScript web, React Native mobile) deployed to GCP Cloud Run via GitHub Actions with Pulumi-managed infrastructure.
- Diagnosed and fixed an upload defect where the client accepted files the API rejected (10MB UI vs. 5MB server cap) and silently stored blank HEIC images; shipped client-side validation and in-browser HEIC-to-JPEG conversion.
- Own the React Native messaging feature; traced a broken media rollout to un-applied Pulumi state and reported the root cause and remaining cutover steps to the engineering lead.

### IT Analyst Intern

May 2026 – Present

*Parker Hannifin*

*Jeffersonville, IN*

- Troubleshoot endpoint hardware, network, account access, and enterprise applications for on-site manufacturing teams, isolating root causes and documenting repeatable fixes.
- Write PowerShell, Batch, Bash, and SQL scripts to automate routine diagnostics, endpoint configuration, and data lookups, cutting repetitive manual steps.

### Technical Lead, Digital Operations

2024 – Present

*Alnur Mosque Islamic Center*

*Louisville, KY*

- Designed and shipped the organization's 11-page public website (semantic HTML/CSS, no framework) on GitHub Pages: programs, events, online booking, donations, and bilingual English/Spanish resources.
- Built a zero-developer publishing pipeline (Python/Pillow + GitHub Actions): volunteers drop photos in a folder and CI deduplicates by SHA-256, corrects EXIF orientation, re-encodes to web sizes, and regenerates the gallery manifest on push.
- Migrated the organization to Google Workspace and authored a Microsoft 365 Nonprofit / Entra ID identity plan; lead livestreaming and A/V for weekly educational programs.

### Software Developer / Internal Tools Developer

Jul. 2022 – Sep. 2023

*King iPhones*

*Louisville, KY*

- Built internal inventory, pricing, and sales-tracking tools in JavaScript and Google Apps Script, plus an Electron/Node.js batch IMEI lookup workflow, replacing manual spreadsheet processes and reducing tracking errors.

## PROJECTS

### Argument Knowledge Base (RAG) | *Python, NLP, LLM Orchestration, REST APIs*

2026

- Built an end-to-end NLP pipeline transforming 32 multi-hour audio debates into a structured, citation-verified RAG knowledge base of 731 canonical arguments and 893 paired rebuttals (1,100+ citations) for LLM retrieval.
- Engineered a two-level fact-verification system validating 712 citations against 4 primary-source APIs, dropping unverifiable claims with full audit logging; orchestrated 8–12 parallel LLM sub-agents across an idempotent, checkpointed pipeline over ~1,000 records.

### School Operations Platform | *TypeScript, Next.js, Supabase (Postgres), Vercel*

2026

- Designed, built, and launched a production bilingual (English/Arabic, full RTL) mobile-first school-operations platform serving 80 active users across admin, teacher, family, and student roles, replacing a year of manual spreadsheets with one PostgreSQL system of record.
- Built a TypeScript serverless backend (106 API routes, 15 Postgres migrations) spanning grading, attendance, scheduling, payroll, and tuition; engineered role-based access with PBKDF2 hashing and Postgres row-level security, covered by 51 regression tests.

### Rumi: Roommate-Matching Platform ([findyourrumi.com](http://findyourrumi.com)) | *TypeScript, Next.js, PostgreSQL, Vercel*

2026

- Founded and shipped a privacy-first roommate-matching platform end to end, with a compliance-gated release architecture and a double opt-in waitlist (hashed single-use tokens, 18+ validation), hardened via self-run security reviews covering CSV injection, rate-limit isolation, and locked-down Postgres/RPC grants.